

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Using a Calculator

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

USING A CALCULATOR · CH1 · L11

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The Map

This lesson collapses Using a Calculator into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Type the whole expression safely

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use brackets for fractions and powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Read standard form display

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Round only at the end

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Estimate the value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 67 website questions, 67 checked solutions, 3 local PDFs read.

01 Type the whole expression safely

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$1.06 \times 10^8 = 106000000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q10

Atlantic surface area

$$1.06 \times 10^8 = 106000000$$

Indian surface area

$$6.86 \times 10^7 = 68600000$$

Finish: (a) 3.74×10^7 , (b) 11

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

10 The table shows information about the surface area of each of the world's oceans.

Ocean	Surface area in square kilometres
Pacific	1.56×10^8
Indian	6.86×10^7
Southern	2.03×10^7
Arctic	1.41×10^7
Atlantic	1.06×10^8

(a) Work out the difference, in square kilometres, between the surface area of the Atlantic Ocean and the surface area of the Indian Ocean. Give your answer in standard form. square kilometres

(2) The surface area of the Pacific Ocean is k times the surface area of the ...

YOUR SOLUTION

02 Use brackets for fractions and powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out $5 \ 6 \ 10 \ 7 \ 10 \ 2 \ 8 \ 10 \ 4 \ 3$
3 . . times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

03 Read standard form display

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out $5 \ 6 \ 10 \ 7 \ 10 \ 2 \ 8 \ 10 \ 4 \ 3$
3 . . times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

04 Round only at the end

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

16 $P = 2a \times c \times d - a = 58.4$ correct to 3 significant figures. $c = 20$ correct to 2 significant figures. $d = 3.6$ correct to 2 significant figures. Work out the upper bound for the value of P . Show your working clearly. Give your answer correct to 2 decimal places.

YOUR SOLUTION

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out $5 \ 6 \ 10 \ 7 \ 10 \ 2 \ 8 \ 10 \ 4 \ 3$
3 . . times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

06 Estimate the value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

16 $P = 2a c d - a = 58.4$ correct to 3 significant figures. $c = 20$ correct to 2 significant figures. $d = 3.6$ correct to 2 significant figures. Work out the upper bound for the value of P . Show your working clearly. Give your answer correct to 2 decimal places.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Type the whole expression safely
"work out", "show", "hence", a familiar exam phrase	Use brackets for fractions and powers
"work out", "show", "hence", a familiar exam phrase	Read standard form display
"work out", "show", "hence", a familiar exam phrase	Round only at the end
"work out", "show", "hence", a familiar exam phrase	Convert standard form
"work out", "show", "hence", a familiar exam phrase	Estimate the value
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.